

Anh Quoc Nguyen

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Center for Theoretical Biological Physics, Rice University - Northeastern University

Homepage

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Google Scholar



RESEARCH INTEREST

Biophysics, Statistical Physics, Tissue Mechanics,
Tissue Rheology, Cell Collective Behavior

EDUCATION

Ph.D. in Biophysics

Sep 2019 – April 2026

Northeastern University, Boston, MA 02115, USA

Thesis: Deciphering Tissue Mechanical Unjamming, Rheology and Plasticity Driven by Cell–Cell Interaction

Bachelor of Science - Physics, Mathematics

Sep 2015-May 2019

Centre College, Danville, KY 40422, USA

GPA: 3.75/4

PUBLICATIONS

- [1] **Nguyen, Anh Q.**, J. Huang, and D. Bi, “**Origin of yield stress and mechanical plasticity in model biological tissues,**” *Nature Communications*, vol. 16, no. 1, p. 3260, 2025.
- [2] G. Cai, **Nguyen, Anh Q.**, Y. Bashirzadeh, S.-S. Lin, D. Bi, and A. P. Liu, “**Compressive stress drives adhesion-dependent unjamming transitions in breast cancer cell migration,**” *Frontiers in Cell and Developmental Biology*, vol. 10, p. 933 042, 2022.
- [3] H. Yang, G. Roy, **Nguyen, Anh Q.**, *et al.*, “**Multicell: geometric learning in multicellular development,**” *Nature Methods*, pp. 1–9, 2025.
- [4] S. Kayal, **Nguyen, Anh Q.**, and D. Bi, “**The Rheology of Living Tissues: From Cells to Organismal Mechanics,**” *Annual Review of Condensed Matter Physics*, vol. 17, 2025.

SELECTED CONFERENCES AND ACADEMIC PROGRAMS

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|--|---------------------------|
| • APS Global Summit 2025 (Oral) | Anaheim - Mar 2025 |
| • APS March Meeting 2024 (Oral) | Minneapolis - March 2024 |
| • International Physics of Living Systems Meeting (Oral & Poster) | Atlanta - Aug 2023 |
| • GRC : Cell Contact and Adhesion (Poster) | New Hampshire - June 2023 |
| • APS March Meeting 2023 (Oral) | Las Vegas - March 2023 |
| • Summer School for Soft Solids and Complex fluids (Poster) | UMass Amherst - June 2022 |
| • APS Division of Fluid Dynamics Meeting (Oral & Poster) | Atlanta - Nov 2018 |

TEACHING EXPERIENCE

- **Lecturer:** Lab for Physics for Life Sciences 1, Lab for Physics for Life Sciences 2
- **Teaching Assistance:** Physics for Life Sciences 1, Electricity and Magnetism 2, Modern Physics, Computational Problem Solving in Physics, Lab for Physics for Life Sciences 1 & 2, Lab for Physics for Engineering 1 & 2

- **Tutor:** Calculus 1,2,3; Linear Algebra; Differential Equation; Classical Mechanics; Probability and Statistics

SKILLS ---

Matlab, Python, Latex